



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal to RANJETH AGENCIES

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

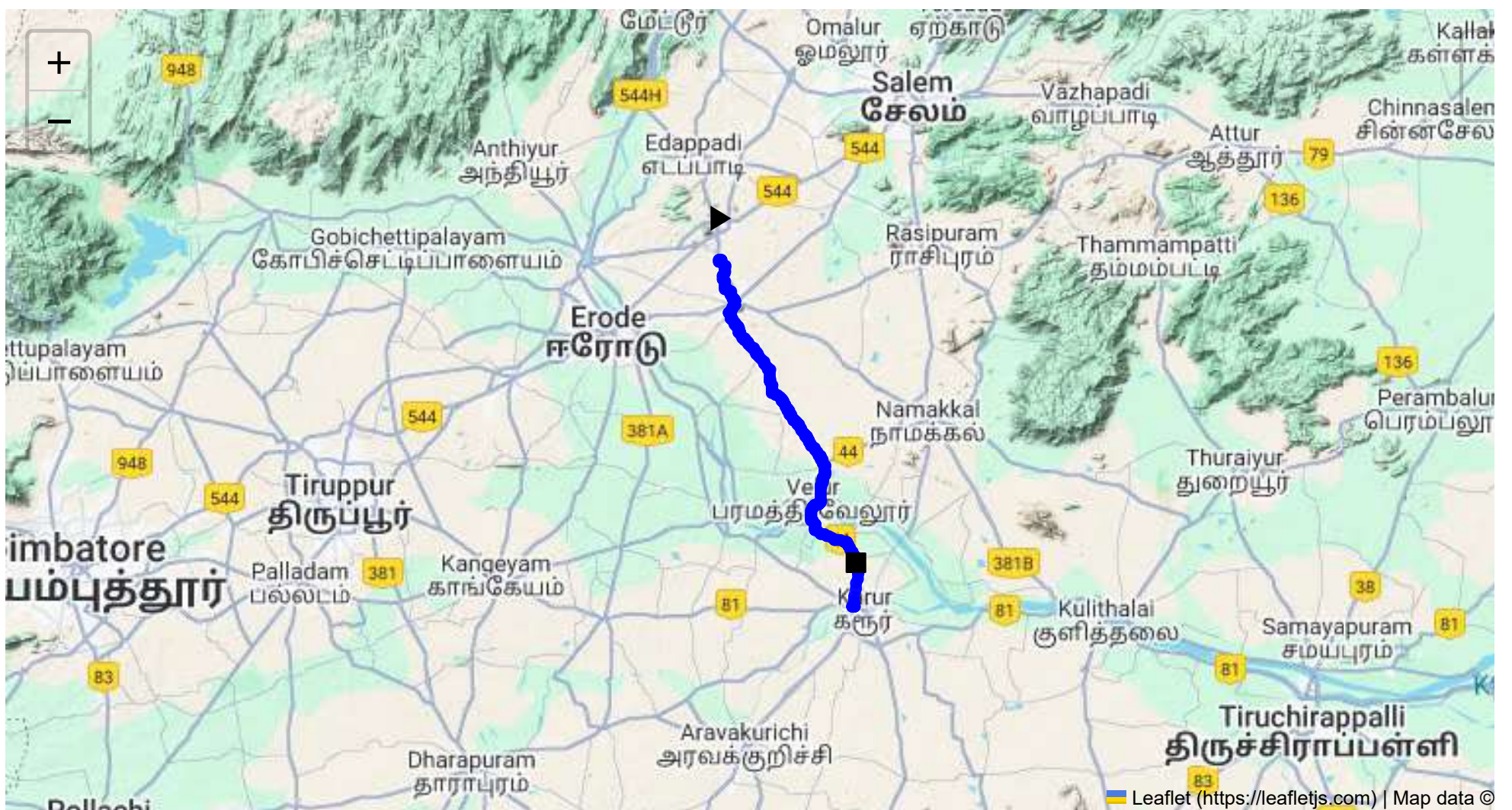
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 64.81 km
Estimated Duration: 1.3 hours
Adjusted Duration (Heavy Vehicle): 1.6 hours
Start: (11.4381, 77.8734)
End: (10.961985, 78.063545)



Welcome to the Journey Risk Management Study

Route Overview

The route from CVQF+23W, Sangagiri, Tamil Nadu to 233, Kovai Main Rd, near Aarunachala hotel, Karur, Tamil Nadu is approximately 64.81 kilometers long. It primarily follows NH544, transitioning to NH44, and then proceeds to local roads towards the destination. The journey typically takes about 1 hour and 29 minutes for heavy vehicles, depending on traffic conditions and vehicle load.

Weather Conditions and Potential Hazards

The region experiences varied weather conditions. The typical weather includes:

- **Summer (March to May):** High temperatures with the possibility of heatwaves. This can lead to tire blowouts if proper tire maintenance is not performed.
- **Monsoon (June to September):** Heavy rainfall can cause waterlogging and slippery roads, increasing the risk of hydroplaning.
- **Winter (November to February):** Generally mild with occasional fog in some areas. Reduced visibility can be a hazard.

Traffic Patterns and Congestion

- **Peak Hours:** Morning (8:00 AM - 10:00 AM) and Evening (5:00 PM - 7:00 PM) are typically congested, especially near urban areas like Salem and Karur.
- **Congestion-Prone Areas:** Near industrial zones and market areas, particularly in towns like Erode and Salem where high vehicle density can lead to slow-moving traffic.

Road Quality and Infrastructure

- **Highways (NH544 and NH44):** Generally well-maintained with clear signage. However, roadworks can occasionally cause diversions.
- **Local Roads:** Some sections may have potholes or be less well-maintained, particularly in rural stretches. Be cautious of sudden changes in road quality.

Alternative Routes

In case of emergencies (e.g., roadblocks, accidents), consider the following alternatives:

- **Via NH544 to NH83:** Redirect through Salem, taking NH83 towards Karur for a more slightly roundabout but feasible route.
- **Local Bypasses:** Utilize smaller local roads as detours, but consider additional time due to potentially lower road quality.

Regulations on Hazardous Material Transport

Transporting hazardous materials requires compliance with local regulations:

- **Permits:** Ensure all required permits are acquired for hazardous materials.
- **Escort Vehicles:** These may be required for specific types of dangerous goods.
- **Operating Hours:** Restrictions may be in place during peak hours to reduce risk.

Historical Incidents

While not frequent, incidents involving trucks include:

- **Rollovers:** Often due to speeding or sharp turns.

- **Spills:** Mainly associated with improper handling or stowage of hazardous materials.

Environmental Considerations

- **Sensitive Areas:** Agricultural regions and water bodies demand cautious driving to prevent contamination.
- **Waste Management:** Dispose of any waste appropriately, especially hazardous substances, in compliance with environmental laws.

Communication Coverage

- **Urban Areas:** Strong coverage along major highways.
- **Rural Stretches:** Potential dead zones, particularly in remote areas away from major cities.

Emergency Response Times

- **Highways:** Approx. 15-20 minutes response time due to good access.
- **Rural Areas:** Can vary from 30-45 minutes depending on location and time of day.

Overall Risk Assessment Summary

The route is considered moderately safe with proper precautions. Key risks include weather-related challenges during the monsoon, traffic congestion near urban centers, and road quality degradation in rural areas. Overall, proactive planning and adherence to regulations ensure a smoother, safer journey for transporting heavy vehicles with hazardous materials. Safety measures like speed regulation, regular vehicle maintenance checks, and effective communication systems are paramount to reducing risks.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	11.38395, 77.89480	15 KM/Hr
1	Roundabout	High	10.96304, 78.06131	15 KM/Hr
2	U-Turn	High	10.9617665, 78.0644854	10 KM/Hr
3	Turn	High	11.43811, 77.87348	15 KM/Hr
4	Turn	High	11.43968, 77.87345	15 KM/Hr
5	Blind Spot	Blind Spot	11.44029, 77.87544	10 KM/Hr
6	Turn	High	11.37868, 77.89498	15 KM/Hr
7	Turn	High	11.37850, 77.89453	15 KM/Hr
8	Turn	Medium	11.16764, 78.01565	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
9	Turn	High	11.15573, 78.02176	15 KM/Hr
10	Turn	High	11.14545, 78.02208	15 KM/Hr
11	Turn	High	11.14480, 78.02291	15 KM/Hr
12	Turn	High	10.96177, 78.06449	15 KM/Hr
13	Blind Spot	Blind Spot	10.96169, 78.06446	10 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
0	hospital	Tiruchengode, Goverment Hospital	11.3903328, 77.8920627	30 km/h	Medium
1	hospital	SPM Medical Centre,Tiruchengode	11.3881331, 77.8931963	30 km/h	Medium
5	hospital	T.C.A Hospital Tiruchengode	11.3791885, 77.8965774	30 km/h	Medium
6	clinic	Kongu Nursing Home	11.3783065, 77.8961134	30 km/h	Medium
7	hospital	Soorya Multispecialty Hospital	11.3786429, 77.8931912	30 km/h	Medium
8	hospital	Tiruchengode Government Hospital	11.37645, 77.89426	30 km/h	Medium
9	hospital	Krishna Hospital, Namakkal	11.3754811, 77.8931817	30 km/h	Medium
10	hospital	Tirukumaran Hospitals	11.3782118, 77.8914933	30 km/h	Medium
13	police	கரட்டுப்பாளையம் காவல் நிலையம்	11.357586, 77.8922539	30 km/h	Medium
14	hospital	Goverment Hospital, Nallur	11.2679698, 77.946863	30 km/h	Medium
16	hospital	Goverment Hospital	11.156647, 78.0213586	30 km/h	Medium
18	hospital	Government Hospital, Manmangalam	11.0275504, 78.0646437	30 km/h	Medium
19	hospital	Abishek Ortho Center	10.9628788, 78.0600715	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
20	hospital	Abishek Ortho Center, Karur	10.9624485, 78.059568	30 km/h	Medium
21	clinic	Senthil Nursing Home, Karur	10.9620158, 78.0643985	30 km/h	Medium
22	hospital	Apollo Hospital	10.96044, 78.064468	30 km/h	Medium
23	hospital	Murugesan Hospital	10.9619995, 78.0643134	30 km/h	Medium
24	hospital	Senthil's Emergency Care	10.9619826, 78.0643823	30 km/h	Medium
25	hospital	Kumaran Hospital	10.9619213, 78.0644924	30 km/h	Medium
26	hospital	Vasan Eye Care Hospital, Karur	10.9608117, 78.0665959	30 km/h	Medium
27	clinic	V.C Nursing Home	10.9605156, 78.0663214	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
2	school	அரசு ஆண்கள் மேல்நிலைப் பள்ளி	11.3850439, 77.8948279	30 km/h	Medium
3	school	அரசு பெண்கள் மேல்நிலைப் பள்ளி	11.3844247, 77.8949053	30 km/h	Medium
4	marketplace	திருச்செங்கோடு தினசரி காய்கறி சந்தை	11.3833608, 77.8970145	30 km/h	Medium
11	school	KSR Educational institution	11.3772013, 77.8908807	30 km/h	Medium
12	school	MDV School	11.3719843, 77.8915244	30 km/h	Medium
15	school	Sivabakkiam Muthusamy Matriculation Higher Secondary School	11.1788177, 78.0061696	30 km/h	Medium
17	school	Goverment Higher Secondary School	11.1511724, 78.0200723	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.38395, 77.89480



Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 10.96304, 78.06131



Risk Type: U-Turn

Risk Level: High

Speed Limit: 10 KM/Hr

Coordinates: 10.9617665, 78.0644854



Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.44029, 77.87544



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.37868, 77.89498



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.37850, 77.89453



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.16764, 78.01565



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.15573, 78.02176



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.14545, 78.02208



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.14480, 78.02291



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 10.96177, 78.06449



Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 10.96169, 78.06446

